

Multithreaded Architectures and Programming on Modern Processors: the Multithreaded Version

Hung-Wei Tseng

Recap: Five implementations

- Which of the following implementations will perform the best on modern pipeline processors?

A

```
inline int popcount(uint64_t x){
    int c=0;
    while(x) {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

B

```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x) {
        c += x & 1;
        x = x >> 1;
        c += x & 1;
        x = x >> 1;
        c += x & 1;
        x = x >> 1;
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

C

```
inline int popcount(uint64_t x) {
    int c = 0;
    int table[16] = {0, 1, 1, 2, 1, 2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
    while(x) {
        c += table[(x & 0xF)];
        x = x >> 4;
    }
    return c;
}
```

D

```
inline int popcount(uint64_t x) {
    int c = 0;
    int table[16] = {0, 1, 1, 2, 1, 2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
    for (uint64_t i = 0; i < 16; i++) {
        c += table[(x & 0xF)];
        x = x >> 4;
    }
    return c;
}
```

E

```
inline int popcount(uint64_t x) {
    int c = 0;
    for (uint64_t i = 0; i < 16; i++) {
        switch((x & 0xF)) {
            case 1: c+=1; break;
            case 2: c+=1; break;
            case 3: c+=2; break;
            case 4: c+=1; break;
            case 5: c+=2; break;
            case 6: c+=2; break;
            case 7: c+=3; break;
            case 8: c+=1; break;
            case 9: c+=2; break;
            case 10: c+=2; break;
            case 11: c+=3; break;
            case 12: c+=2; break;
            case 13: c+=3; break;
            case 14: c+=3; break;
            case 15: c+=4; break;
            default: break;
        }
        x = x >> 4;
    }
    return c;
}
```

Tips of programming on modern processors

- Minimize the critical path operations
 - Don't forget about optimizing cache/memory locality first!
 - Memory latencies are still way longer than any arithmetic instruction
 - Can we use arrays/hash tables instead of lists?
 - Branch can be expensive as pipeline get deeper
 - Sorting
 - Loop unrolling
 - Still need to carefully avoid long latency operations (e.g., mod)
- Since processors have multiple functional units — code must be able to exploit instruction-level parallelism
 - Hide as many instructions as possible under the "critical path"
 - Try to use as many different functional units simultaneously as possible
- Modern processors also have accelerated instructions
- Compiler can do fairly go optimizations, but with limitations

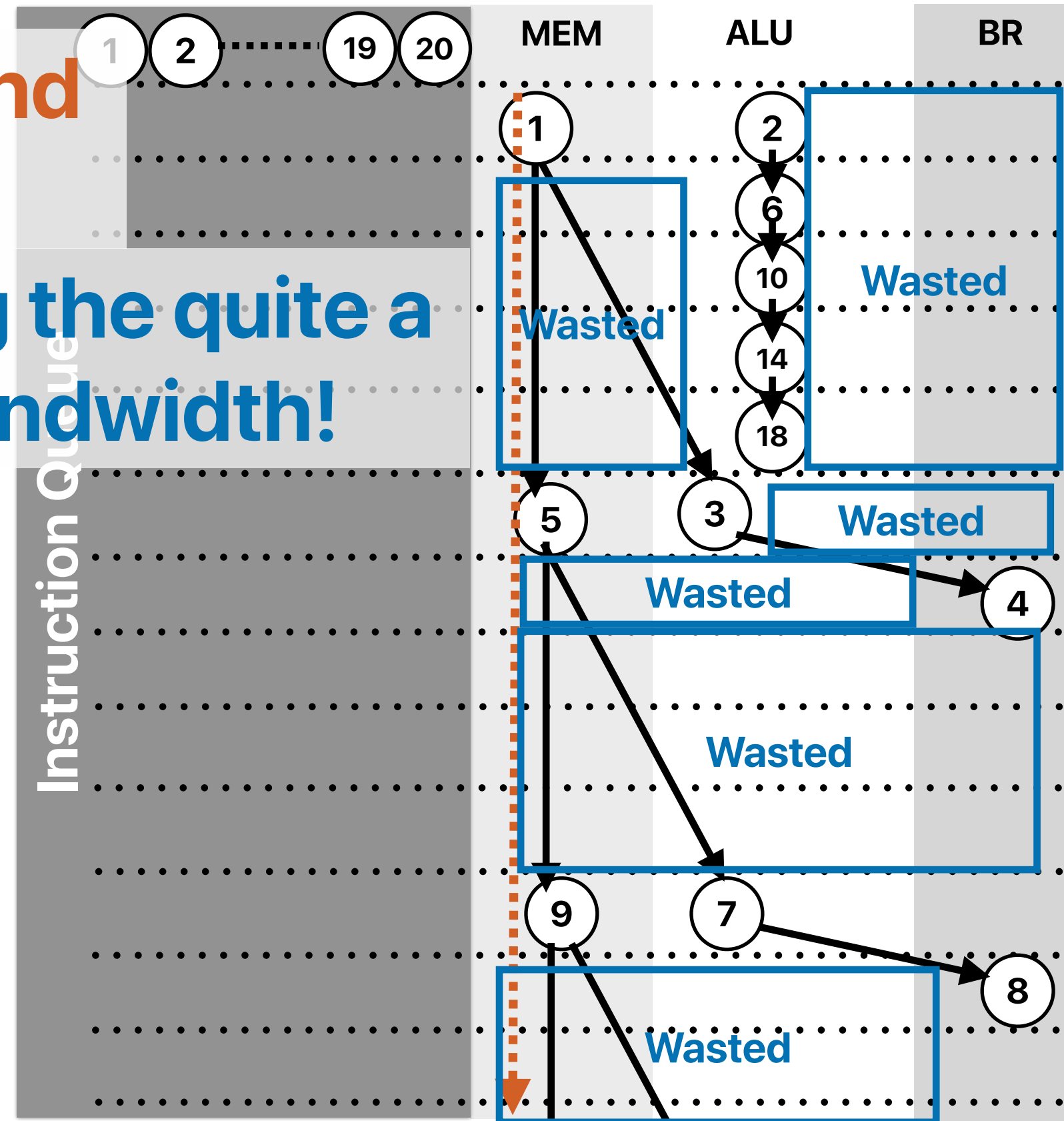
What if we have "unlimited" fetch/issue width — "linked list"

Even unlimited issue width and a perfect cache cannot help!

We're wasting the quite a lot issue bandwidth!

```
do {  
    number_of_nodes++;  
    current = current->next;  
} while ( current != NULL );
```

```
① .L3:      movq      8(%rdi), %rdi
②          addl      $1, %eax
③          testq     %rdi, %rdi
④          jne       .L3
```



Summary of popcounts

	ET	IC	IPC/ILP	# of branches	Branch mis-prediction rate
A	22.21	332 Trillions	2.88	65 Trillions	1.13%
B	12.29	287 Trillions	4.52	17 Trillions	0.04%
C	5.01	102 Trillions	3.95	17 Trillions	0.04%
D	3.73	80 Trillions	4.13	1 Trillions	~0%
E	54.4	173 Trillions	0.61	44 Trillions	18.6%
SSE4.2	1.57	22 Trillions	2.7	1 Trillions	~0%

Best at 4.5

Best performing one at 2.7

Outline

- Parallel architectures
 - Simultaneous multithreading (SMT)
 - Chip multiprocessors (CMP)
- Parallel programming

Parallel architectures

Simultaneous multithreading

Concept: Simultaneous Multithreading (SMT)

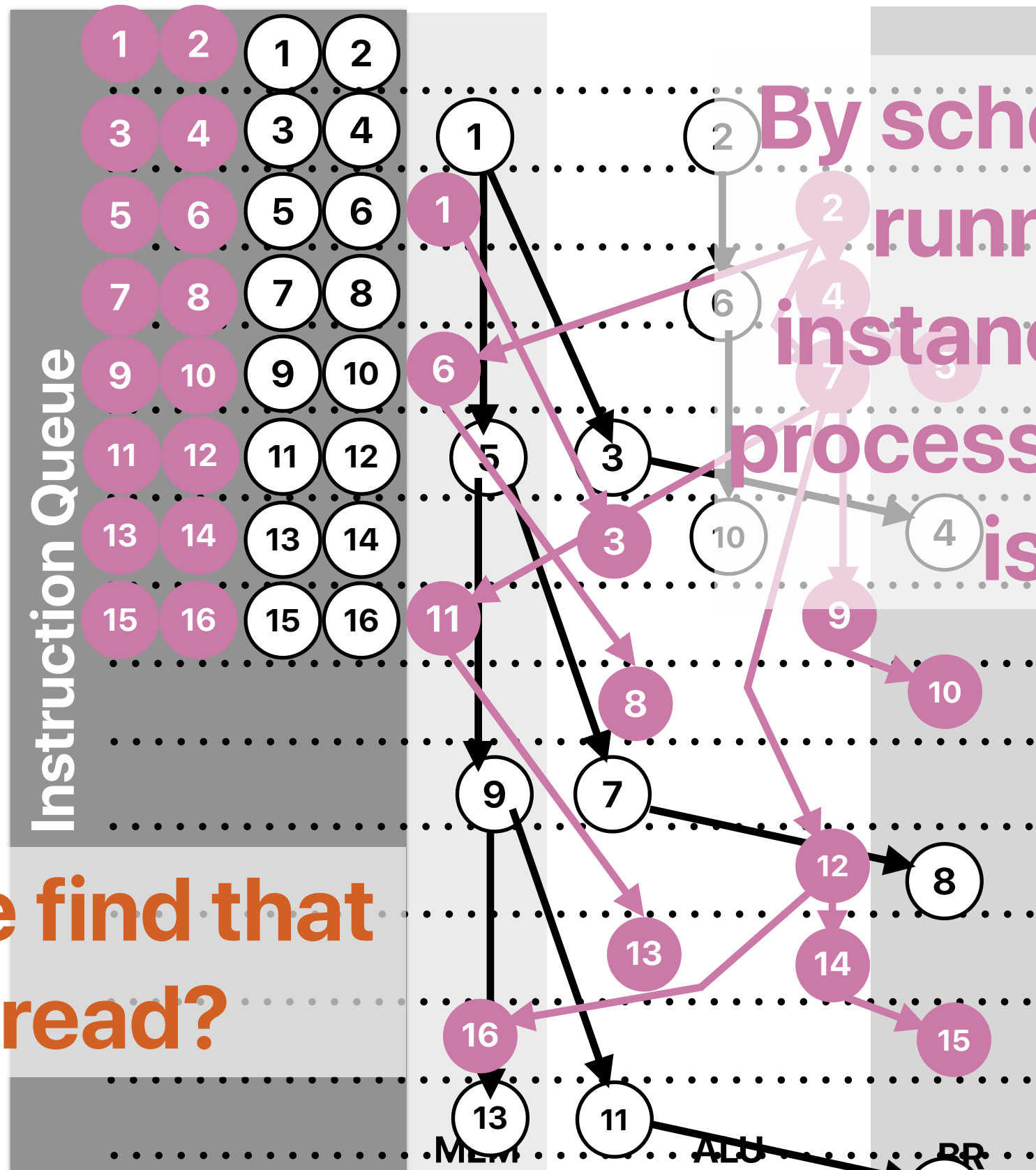
```

① movq    8(%rdi), %rdi
② addl    $1, %eax
③ testq   %rdi, %rdi
④ jne     .L3
⑤ movq    8(%rdi), %rdi
⑥ addl    $1, %eax
⑦ testq   %rdi, %rdi
⑧ jne     .L3
⑨ movq    8(%rdi), %rdi
⑩ addl    $1, %eax
⑪ testq   %rdi, %rdi
⑫ jne     .L3

```

Where can we
"other" the

Where can we find that "other" thread?



By scheduling another running program instance (thread), the processor has 0 wasted issue slots!

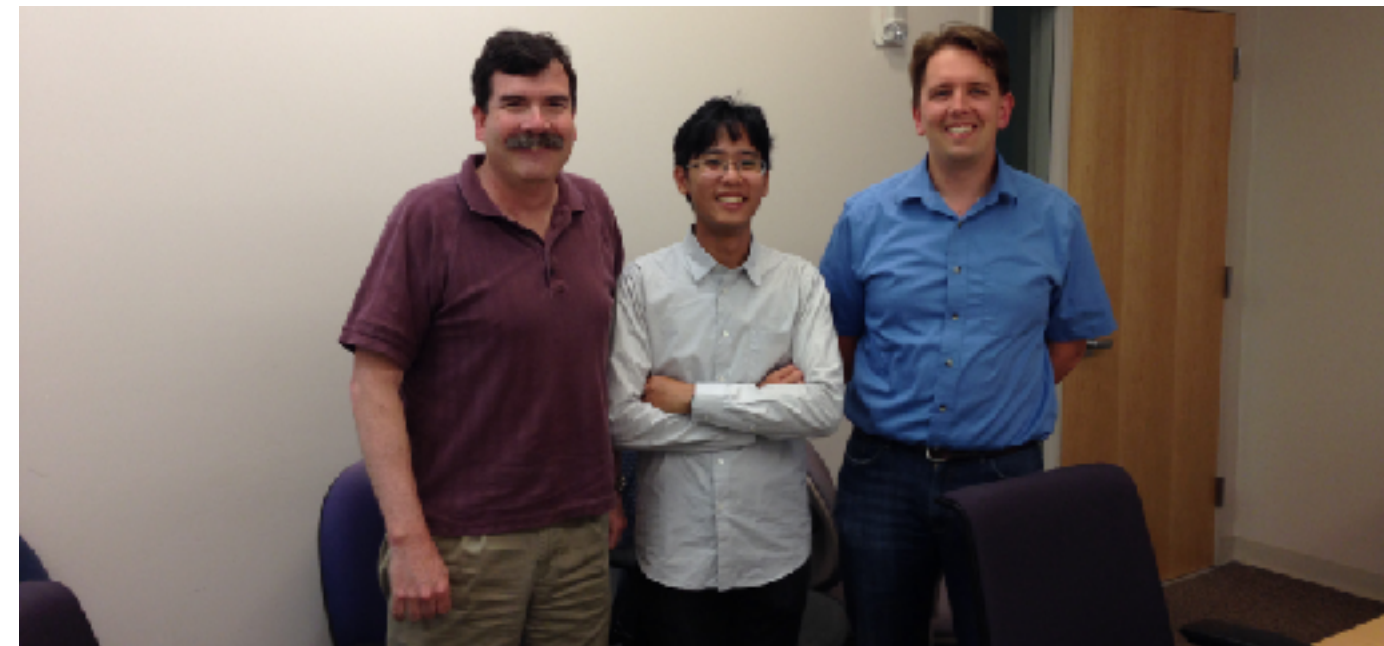
the (thread), the
or has 0 wasted
use slots!

Where to find another "thread"

- Another process/running program forked from a completely different program
- Another process forked/cloned from the current process
 - The forked program cloned the memory content at the forked time
 - The forked program cannot view the memory space of the original process
 - The forked program can perform a subset of tasks from the original process
 - The forked and the original process can exchange data through files or inter-process communication APIs
- A software thread spawned from the current process
 - The spawned thread executes a function instance from the main process to offload computation from the main process
 - The spawned thread shares the memory space with the main process

Simultaneous multithreading

- The processor can schedule instructions from different threads/processes/programs
- Fetch instructions from different threads/processes to fill the not utilized part of pipeline
 - Exploit “thread level parallelism” (TLP) to solve the problem of insufficient ILP in a single thread
 - You need to create an illusion of multiple processors for OSs
- Invented by Dean Tullsen (Now a professor at **UCSD CSE**)



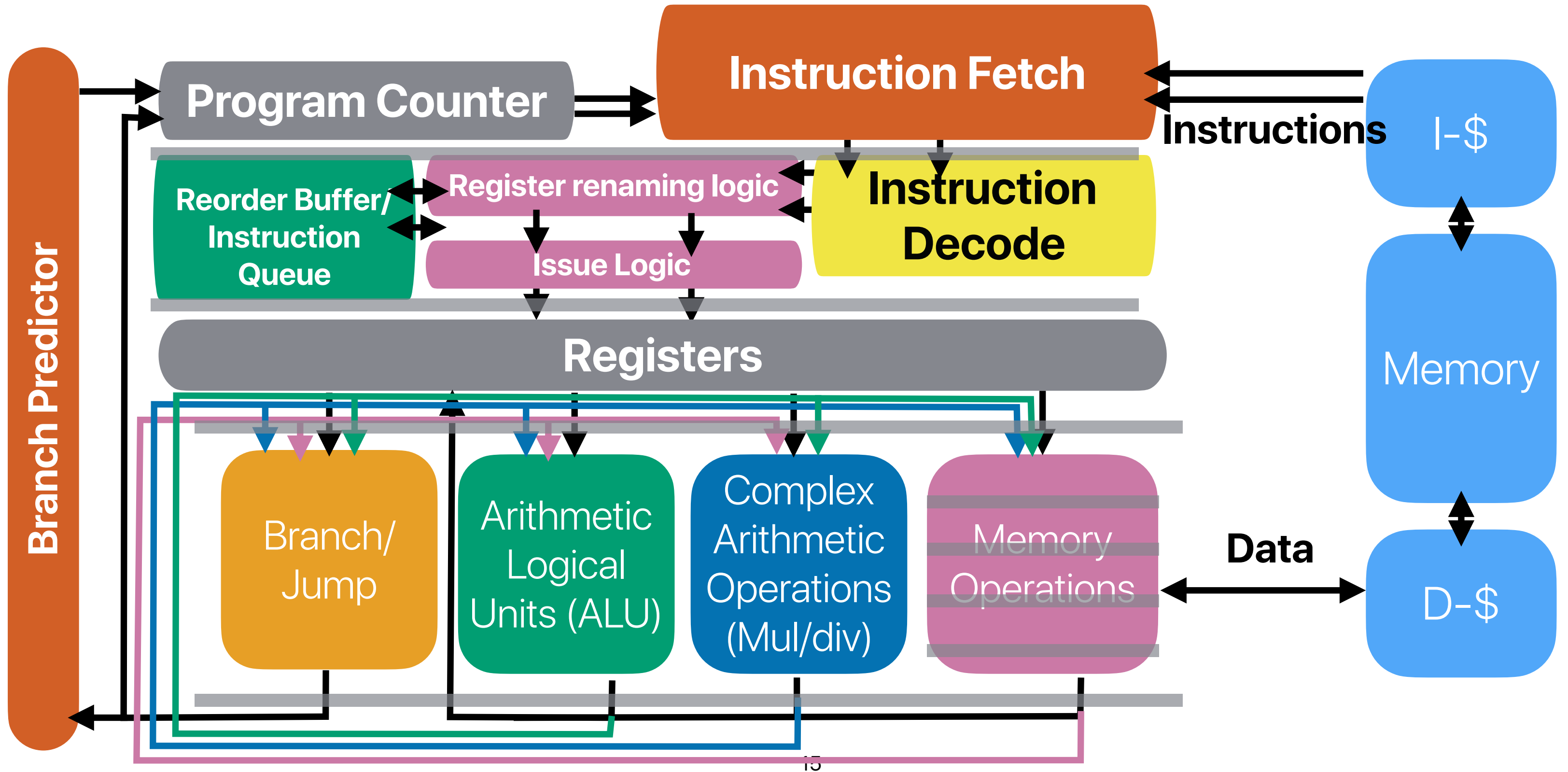
SMT from the user/OS' perspective



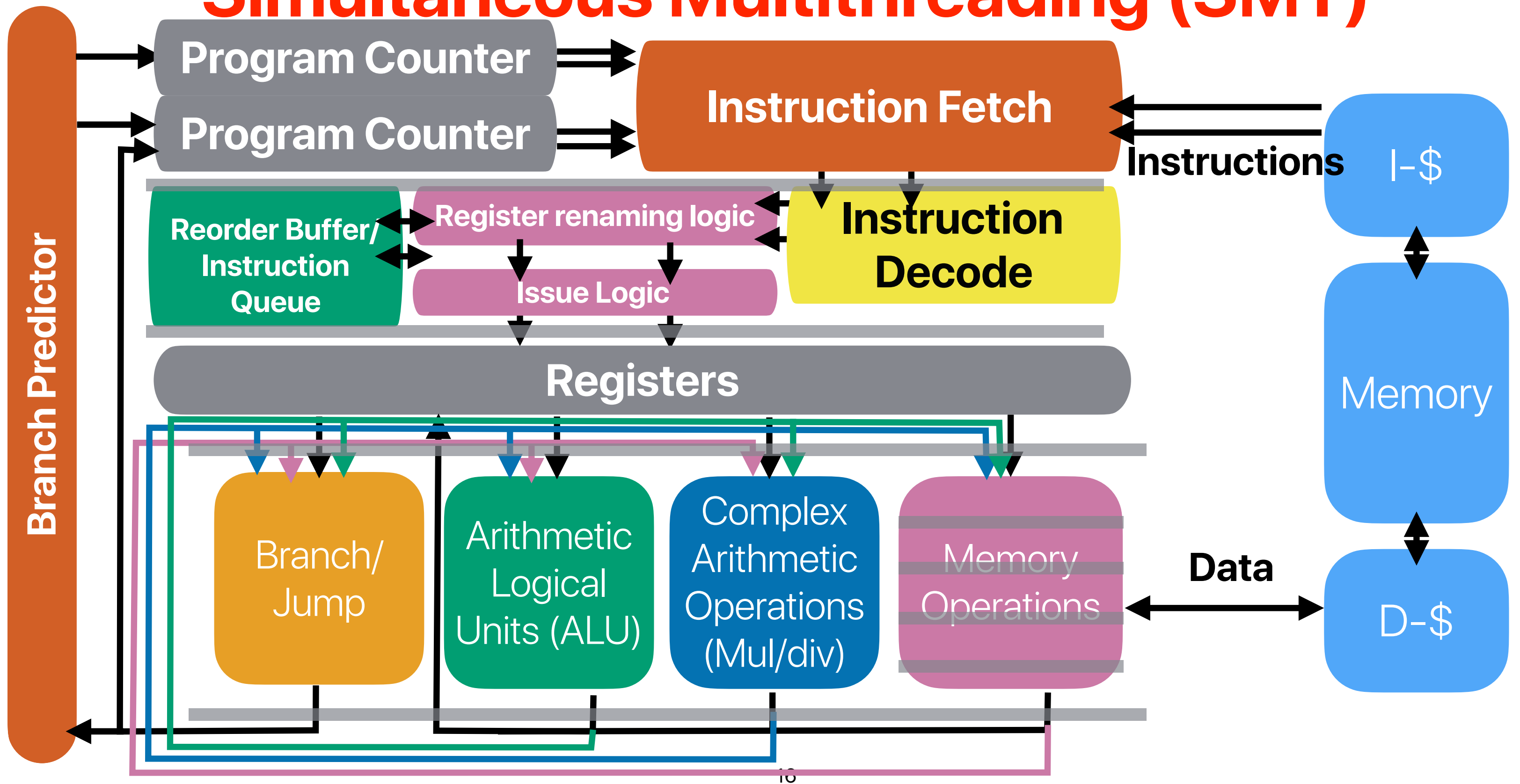
How do we support two running programs in one pipeline?

- We need two program counters
- We need two sets of architectural to physical register mappings
- We do not need
 - Duplicated cache — virtually indexed, physically tagged cache already addressed that
 - Duplicated pipeline functional units — isn't sharing the whole purpose?
 - Duplicated reorder buffer — you simply need to tag which process the instruction belongs to

Recap: Register renaming



Simultaneous Multithreading (SMT)





Pros/Cons of SMT

- How many of the following statements about SMT is/are correct?
 - ① SMT makes processors with deep pipelines more tolerable to mis-predicted branches
 - ② SMT can improve the throughput of a single-threaded application
 - ③ SMT processors can better utilize hardware during cache misses comparing with superscalar processors with the same issue width
 - ④ SMT processors can have higher cache miss rates comparing with superscalar processors with the same cache sizes when executing the same set of applications.

A. 0
B. 1
C. 2
D. 3
E. 4



Pros/Cons of SMT

- How many of the following statements about SMT is/are correct?

- ① ✓ SMT makes processors with deep pipelines more tolerable to mis-predicted branches
We can execute from other threads/contexts instead of the current one
hurt, b/c you are sharing resource with other threads.
- ② SMT can ~~improve~~ the throughput of a single-threaded application
- ③ ✓ SMT processors can better utilize hardware during cache misses comparing with superscalar processors with the same issue width
We can execute from other threads/contexts instead of the current one
- ④ ✓ SMT processors can have higher cache miss rates comparing with superscalar processors with the same cache sizes when executing the same set of applications.
b/c we're sharing the cache

A. 0

B. 1

C. 2

D. 3

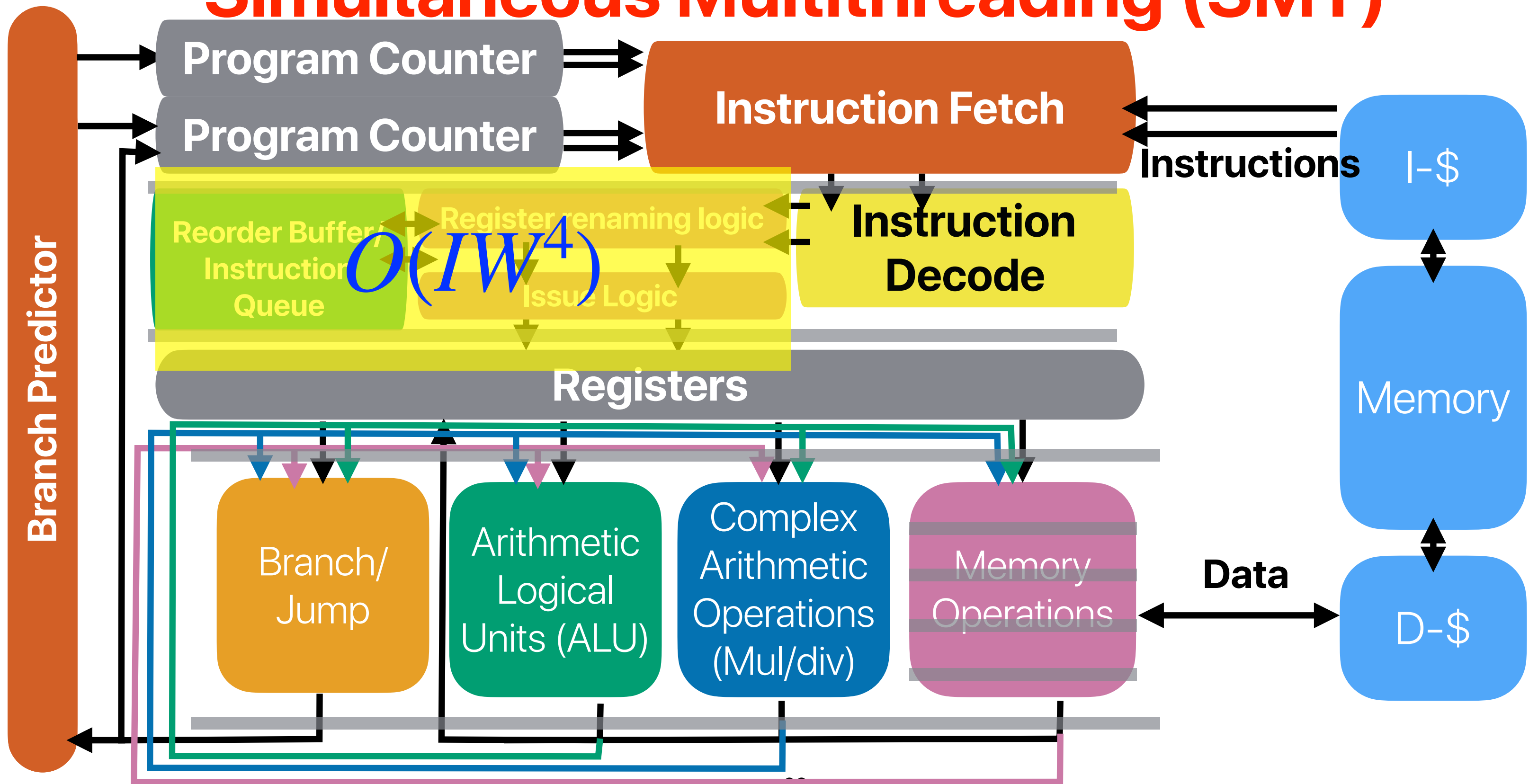
E. 4

```
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Byte Order: Little Endian
Address sizes: 39 bits physical, 48 bits virtual
CPU(s): 8
On-line CPU(s) list: 0-7
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s): 1
NUMA node(s): 1
Vendor ID: GenuineIntel
CPU family: 6
Model: 151
Model name: 12th Gen Intel(R) Core(TM) i3-12100F
Stepping: 5
CPU MHz: 3300.000
CPU max MHz: 5500.0000
CPU min MHz: 800.0000
BogoMIPS: 6604.80
Virtualization: VT-x
L1d cache: 192 KiB
L1i cache: 128 KiB
```

SMT in real practice

- Intel HyperThreading (supports up to two threads per core)
 - Intel Pentium 4, Intel Atom, All Intel Core i7s and Core i9s
- AMD RyZen (Zen microarchitecture)
- If you see a processor with "threads" more than "cores", that must be because of SMT!

Simultaneous Multithreading (SMT)



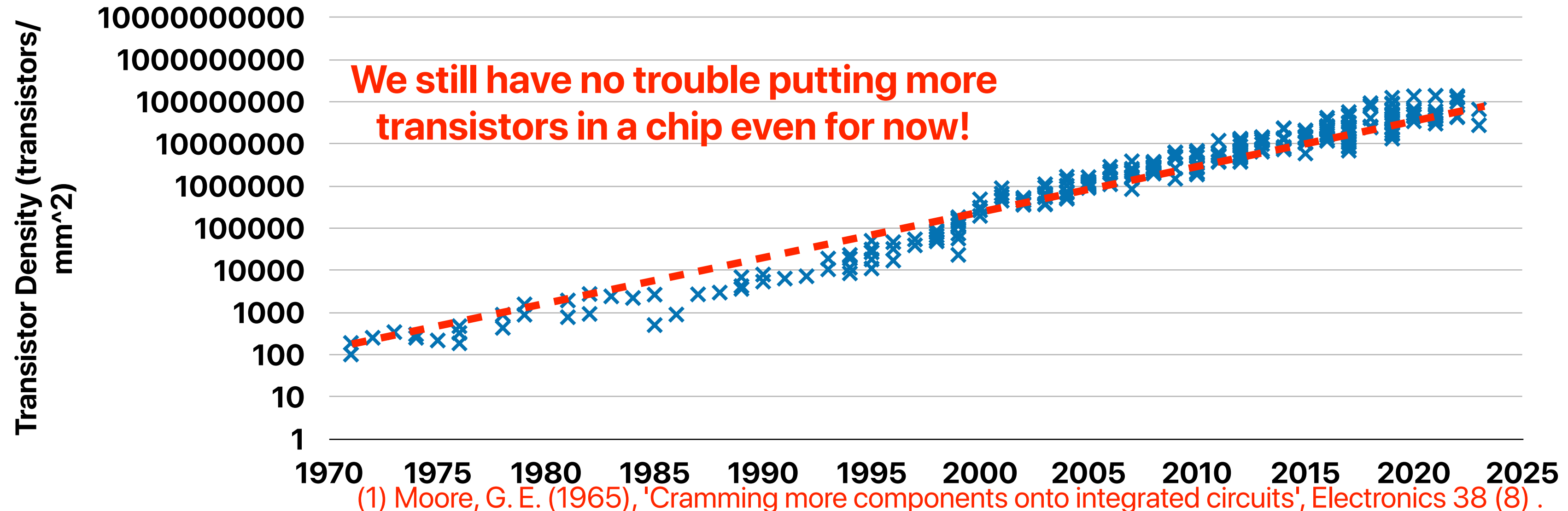
Takeaways: parallel architectures

- SMT processors can better utilize the pipeline resources by allowing simultaneous execution of multiple threads
 - Improved execution throughput
 - May hurt the latency of each thread since we share functional units & cache

Chip-Multiprocessors (CMP) or Multi-core processors

Recap: Moore's Law⁽¹⁾

- The number of transistors we can build in a fixed area of silicon doubles every 12 ~ 24 months.
- Moore's Law "was" the most important driver for historic CPU performance gains



Transistor Counts

Microarchitecture	Transistor Count	Issue-width	Year
Alder Lake	325 M	5x ALU, 7x Memory	2021
Coffee Lake	217 M	4x ALU, 4x Memory	2017
Sandy Bridge	290 M	3x ALU, 3x Memory	2011
Nehalem	182.75 M	3x ALU, 3x Memory	2008



How many transistors per core on Coffee Lake?



The Coffee Lake processor has 217 million transistors per core. It is manufactured using Intel's second 14 nm process. Coffee Lake processors introduced i5 and i7 CPUs featuring six cores (along with hyper-threading in the case of the i7) and no hyperthreading.

The transistor count per core on Coffee Lake is lower than that of some other modern processors, such as the AMD Ryzen 5 5600X. However, Coffee Lake still offers good performance, thanks to its high clock speeds and efficient architecture.

Here is a table of the transistor counts per core for some other modern processors:

Processor	Transistors per core
Coffee Lake	217 million
Ryzen 5 5600X	390 million
Ryzen 9 5950X	52 million

Nehalem Alder Lake
6-issue 12-issue

Recap: do we really need very wide issue processors?

	ET	IC	IPC/ILP	# of branches	Branch mis-prediction rate
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One powerful core or two "good enough" cores?

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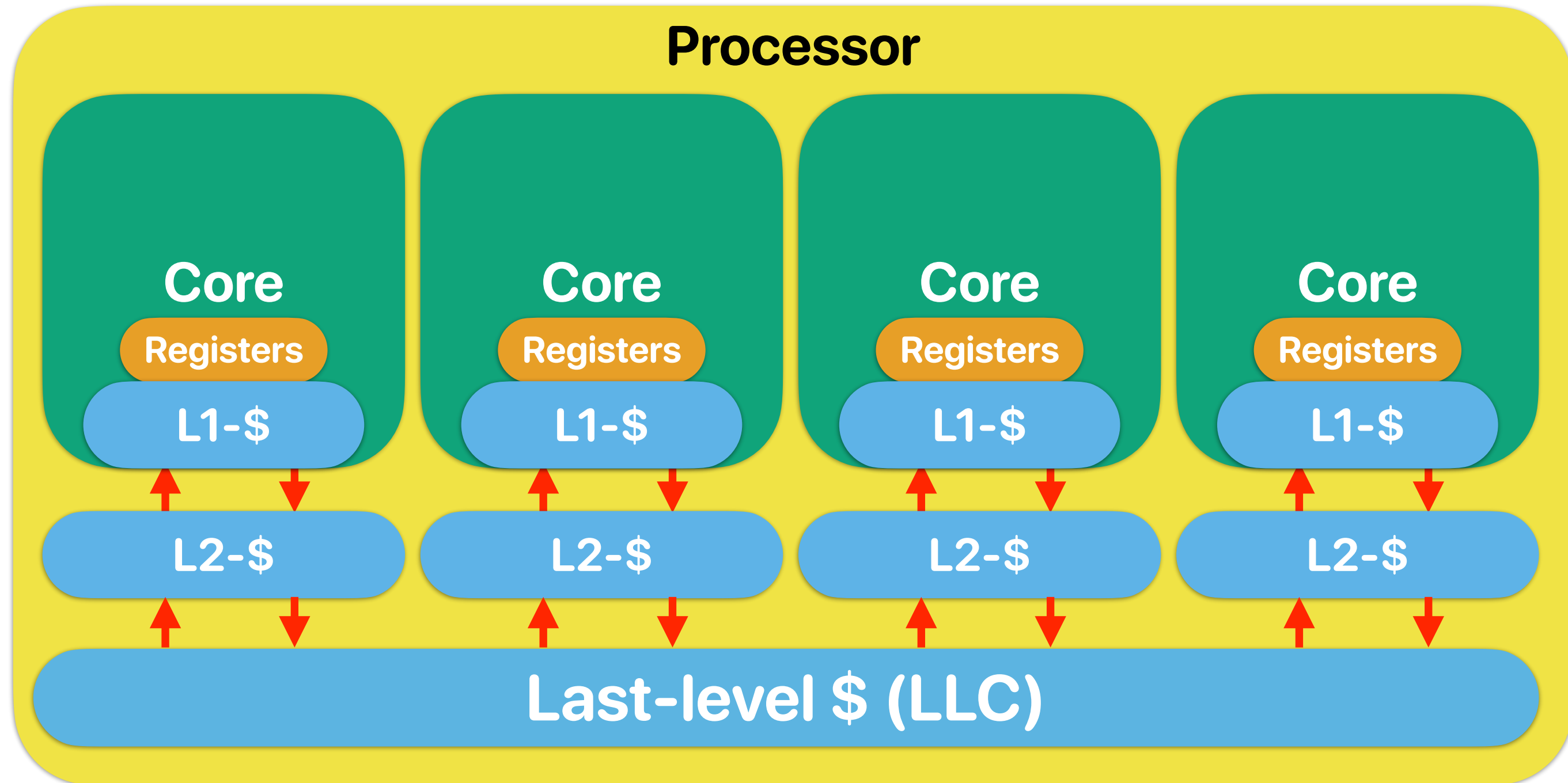
2x 3-issue ALUs Nehalem

Nehalem Alder Lake Nehalem
6-issue 12-issue 6-issue

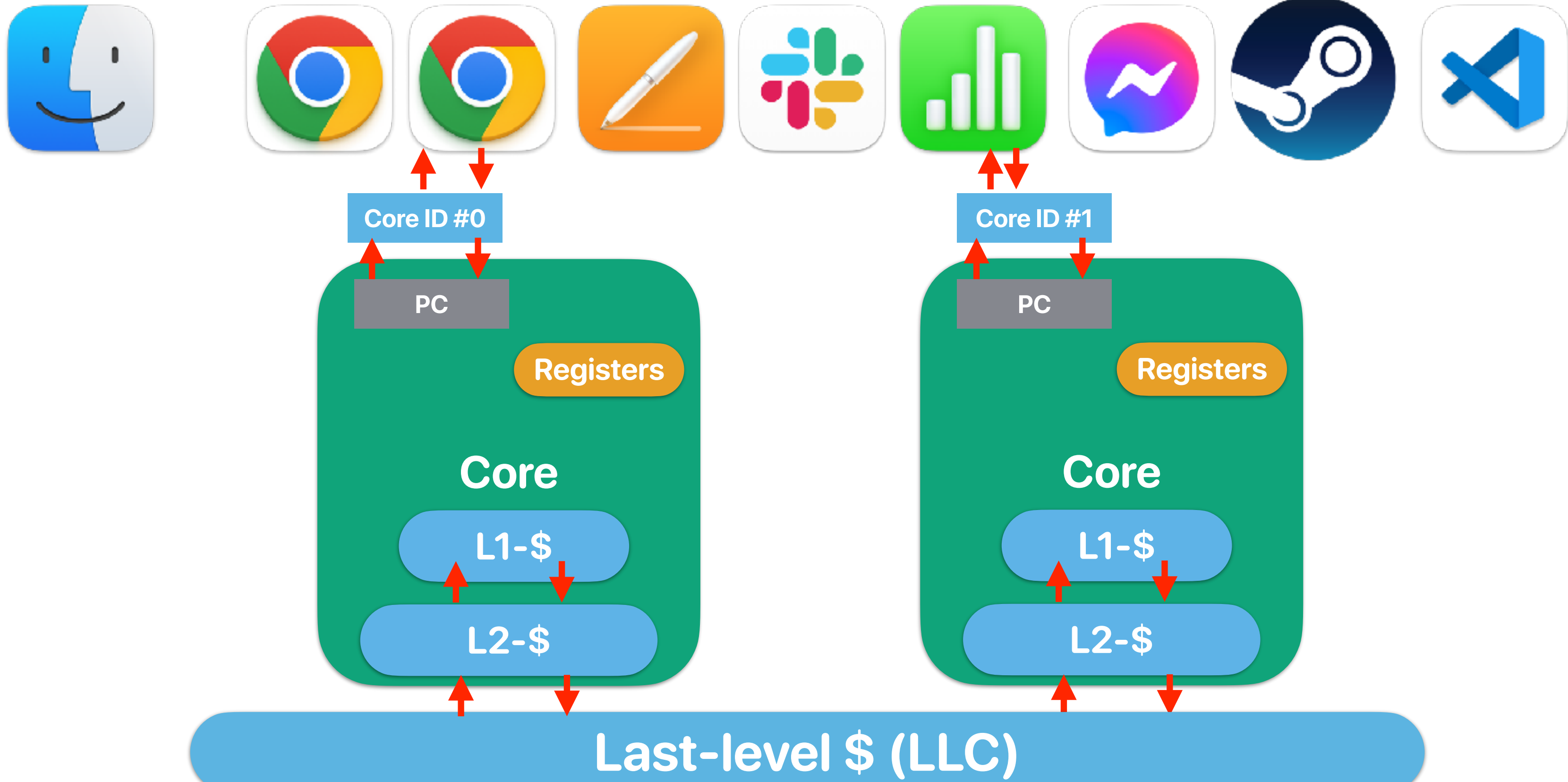
1x 5-issue ALUs Alder Lake

Based on https://en.wikipedia.org/wiki/Transistor_count

Concept of CMP



CMP from the user/OS' perspective





SMT v.s. CMP

- An SMT processor is basically a SuperScalar processor with multiple instruction front-end. Assume within the same chip area, we can build an SMT processor supporting 4 threads, with 6-issue pipeline, 64KB cache or — a CMP with 4x 2-issue pipeline & 16KB cache in each core. Please identify how many of the following statements are/is correct when running programs on these processors.
 - ① If we are just running one program in the system, the program will perform better on an SMT processor
 - ② If we are running 4 applications simultaneously, the cache miss rates will be higher in the SMT processor
 - ③ If we are running 4 applications simultaneously, the branch mis-prediction will be higher in the SMT processor
 - ④ If we are running one program with 4 parallel threads, the cache miss rates will be higher in the SMT processor
 - ⑤ If we are running one program with 4 parallel threads simultaneously, the branch mis-prediction will be longer in the SMT processor

A. 1
B. 2
C. 3
D. 4
E. 5



SMT v.s. CMP

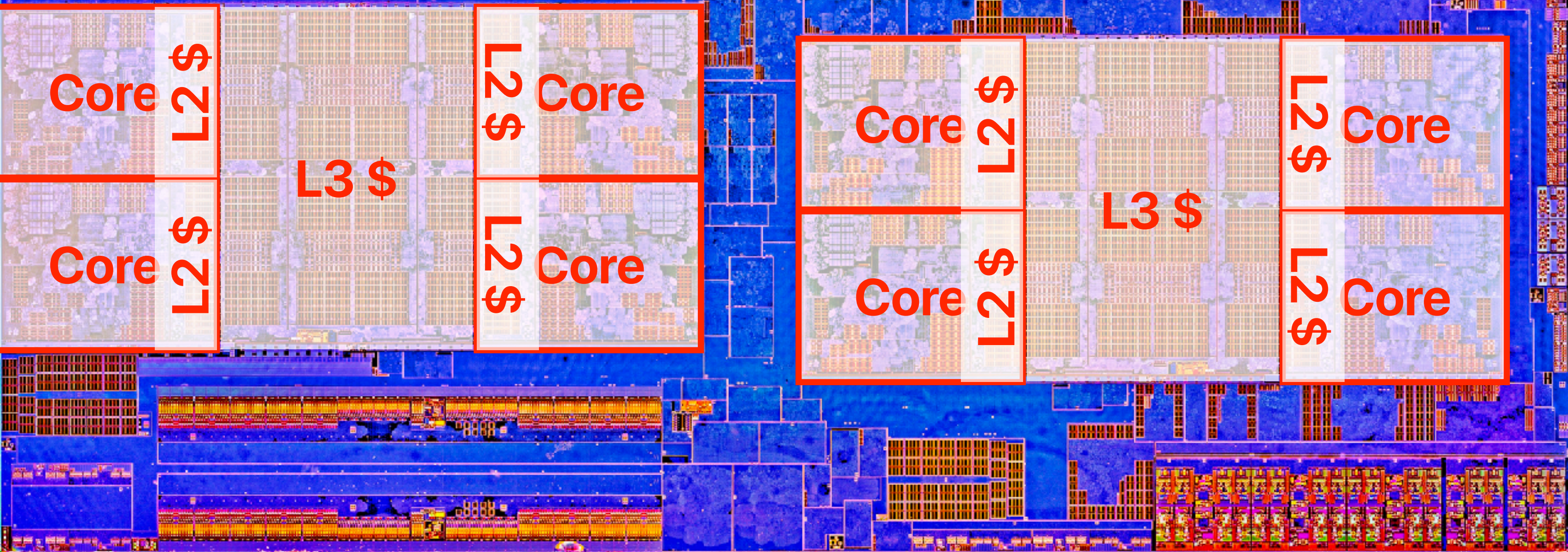
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 - ① If we are just running one program in the system, the program will perform better on an SMT processor — **you have more resources for the program**
 - ② If we are running 4 applications simultaneously, the cache miss rates will be higher in the SMT processor
 - ③ If we are running 4 applications simultaneously, the branch mis-prediction will be higher in the SMT processor — **it depends!**
 - ④ If we are running one program with 4 parallel threads, the cache miss rates will be higher in the SMT processor — **it depends!**
 - ⑤ If we are running one program with 4 parallel threads simultaneously, the branch mis-prediction will be longer in the SMT processor — **it depends!**
- A. 1 **There is no clear win on each —**
B. 2 **why not having both?**
C. 3
D. 4
E. 5


```
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
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```


Modern processors have both CMP/SMT



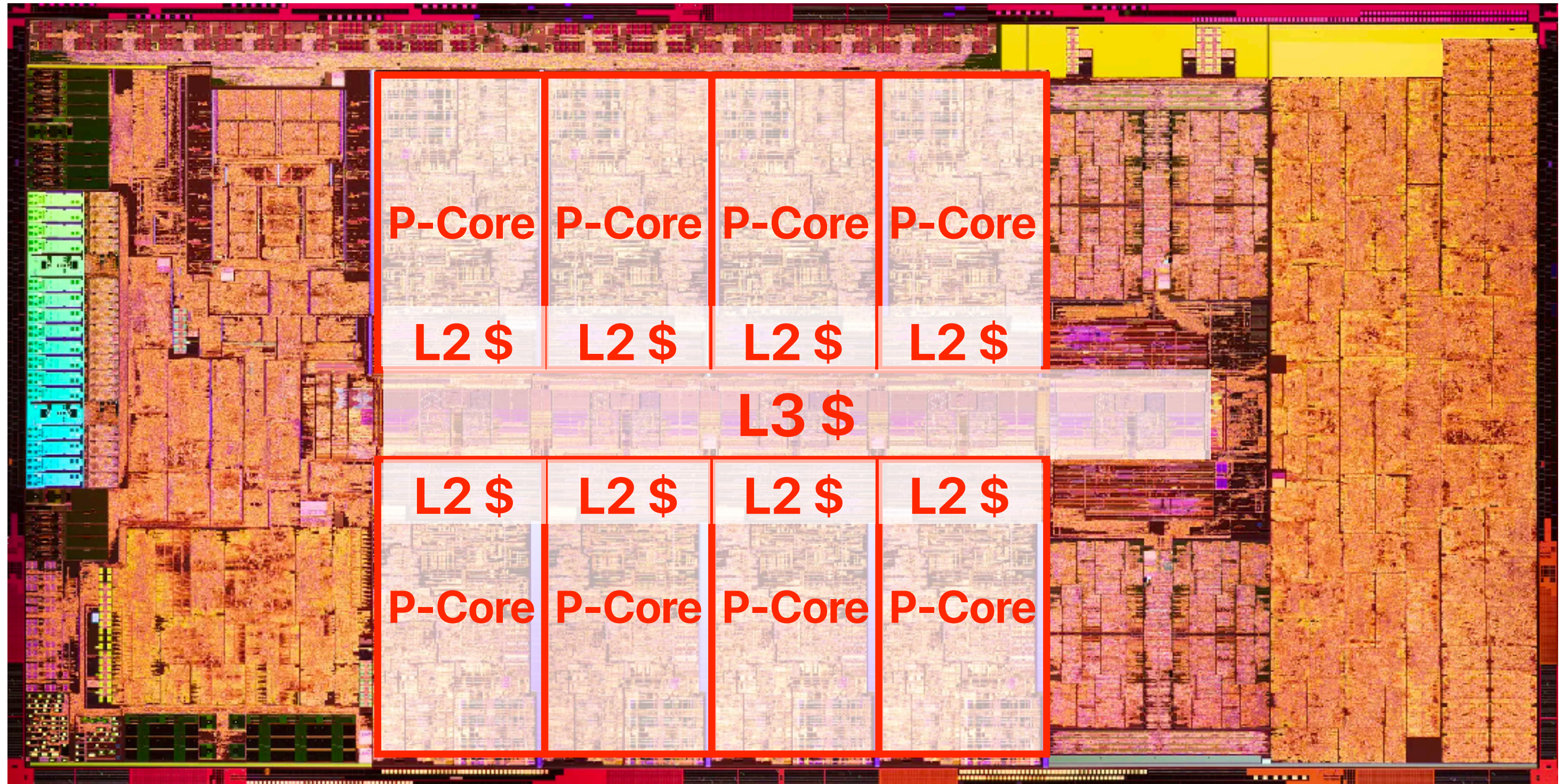
AMD RyZen Processor



AMD

RYZEN

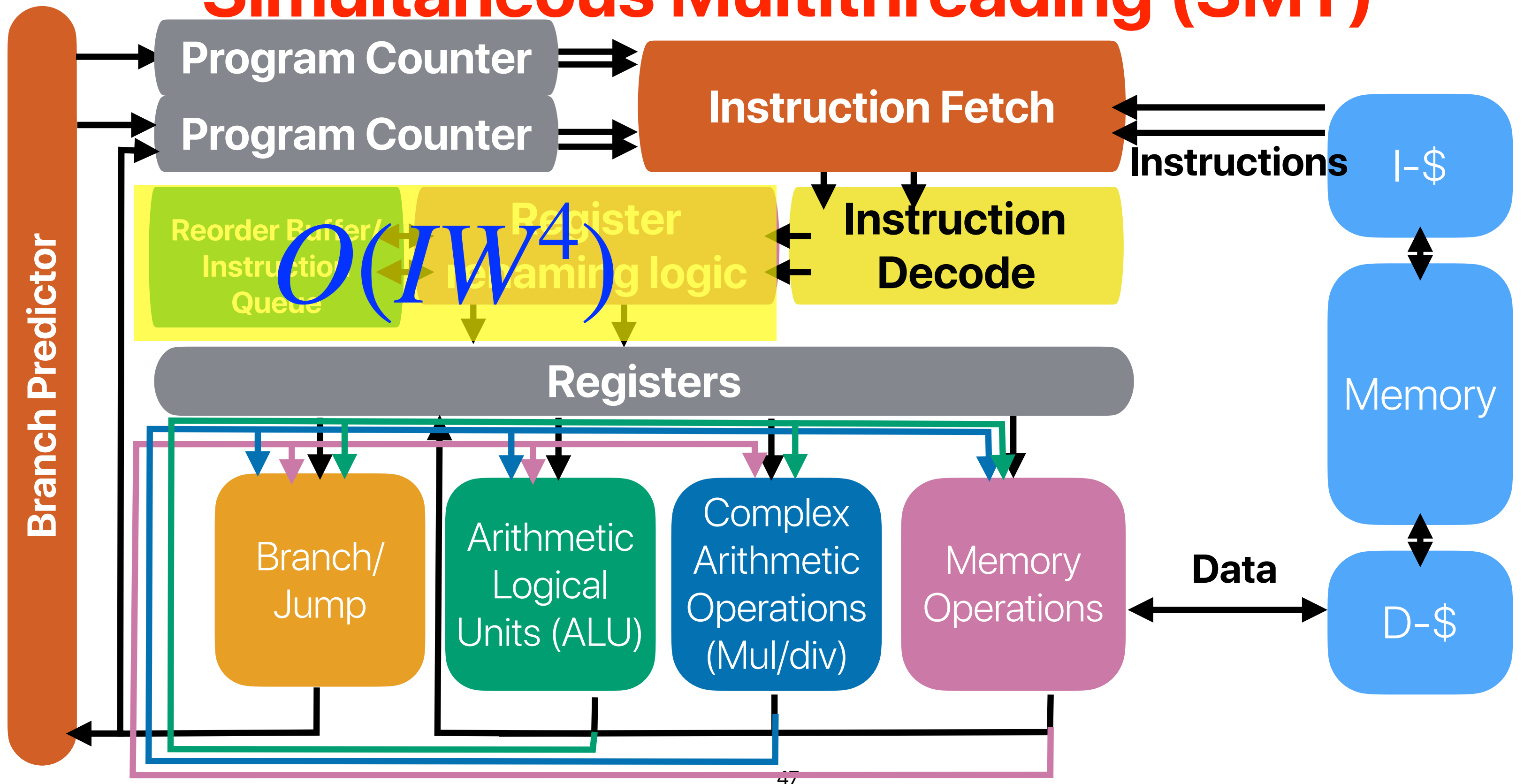
Intel Alder Lake



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- A. 1 **There is no clear win on each —**
B. 2 **why not having both?**
C. 3
D. 4 **The only thing we know for sure**
E. 5 **— if we don't parallel the program, it won't get any faster on SMT/CMP**

Simultaneous Multithreading (SMT)



Takeaways: parallel architectures

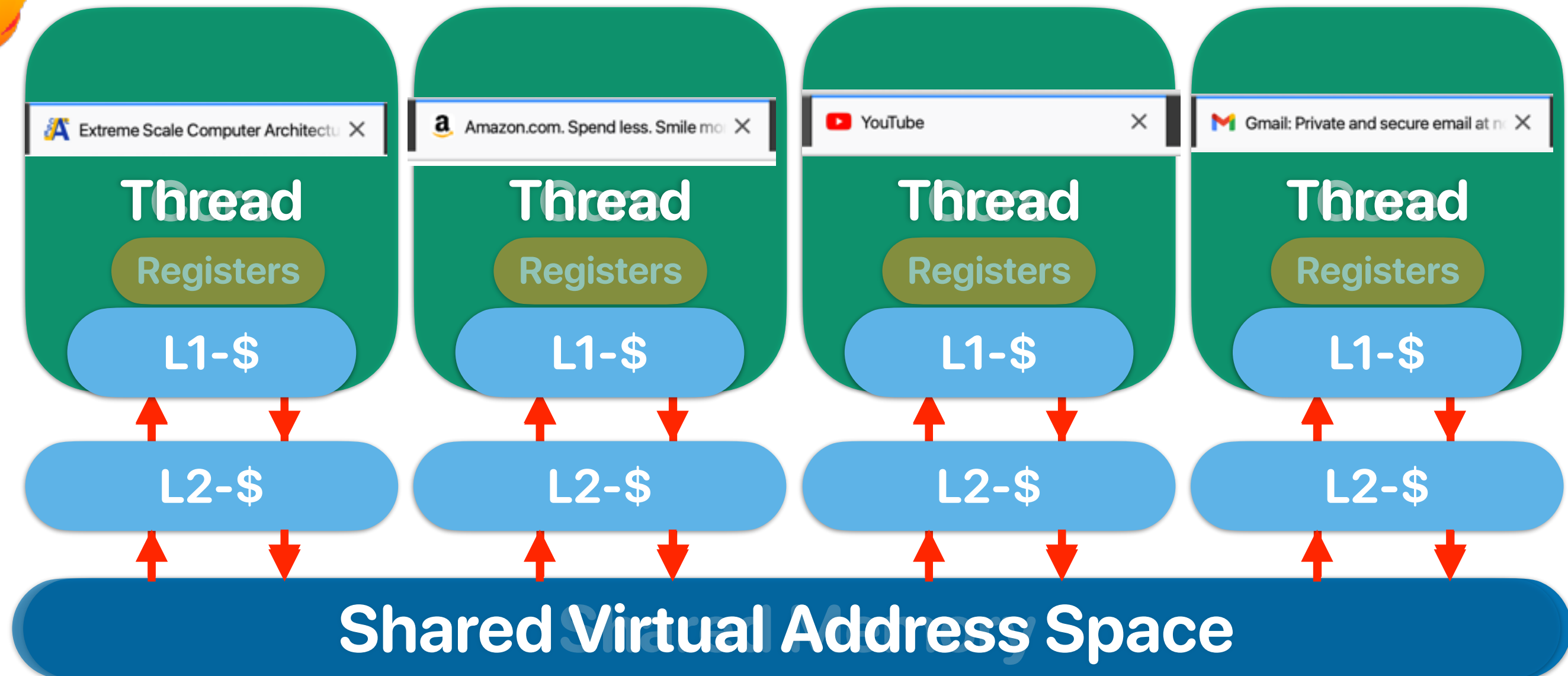
- SMT processors can better utilize the pipeline resources by allowing simultaneous execution of multiple threads
 - Improved execution throughput
 - May hurt the latency of each thread since we share functional units & cache
- CMP provides more processor cores for parallel threads or multiprogrammed environments
 - More isolated, protected pipeline
 - No flexibility if the running program needs more resource

Parallel Programming & Architectural Supports for Parallel Programming

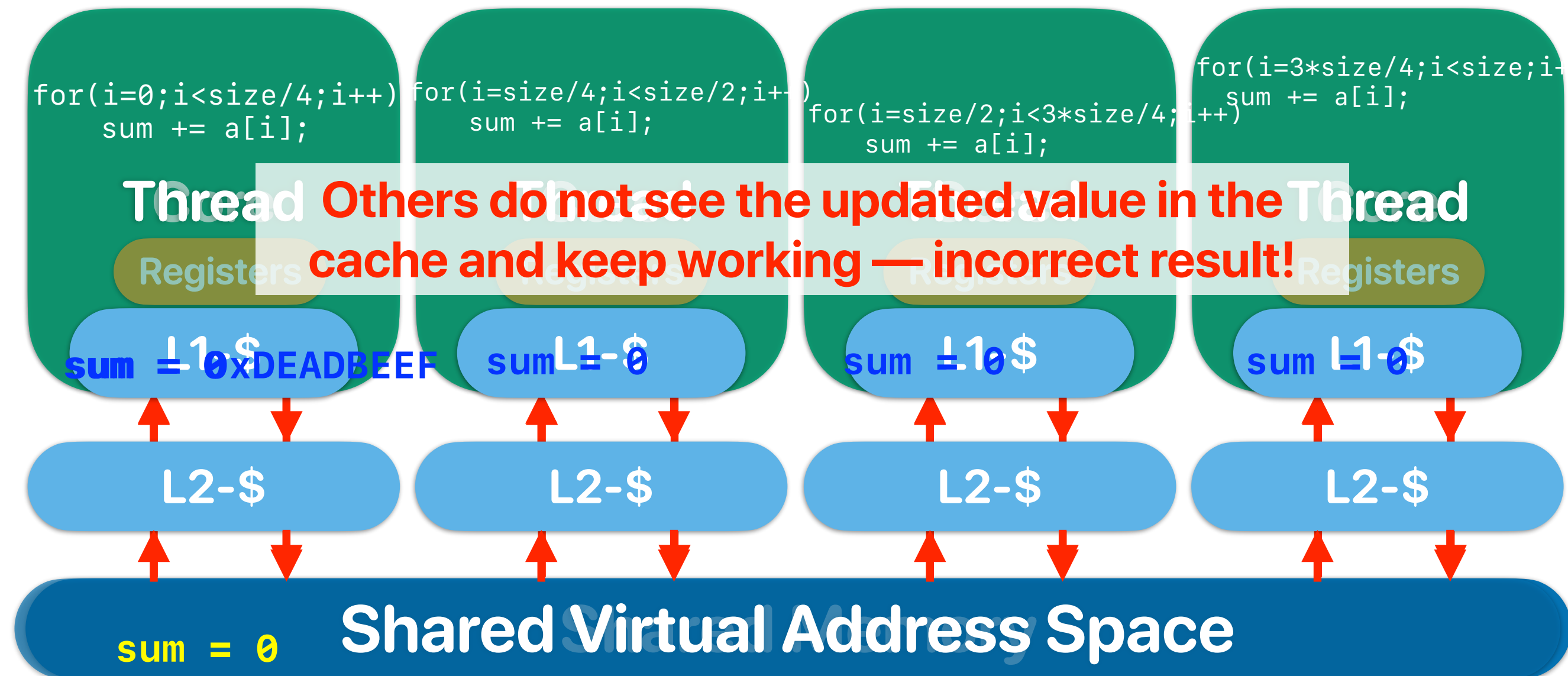
Parallel programming

- To exploit parallelism you need to break your computation into multiple “processes” or multiple “threads”
- Processes (in OS/software systems)
 - Separate programs actually running (not sitting idle) on your computer at the same time.
 - Each process will have its own virtual memory space and you need explicitly exchange data using inter-process communication APIs
 - Programming model: MPI, Spark
- Threads (in OS/software systems)
 - Independent portions of your program that can run in parallel
 - All threads share the same virtual memory space
 - Programming model: pthread or openmp
- We will refer to these collectively as “threads”
 - A typical user system might have 1-8 actively running threads.
 - Servers can have more if needed (the sysadmins will hopefully configure it that way)

What software thinks about "multiprogramming" hardware



What software thinks about "multiprogramming" hardware



Coherency & Consistency

- Coherency — Guarantees all processors see the same value for a variable/memory address in the system when the processors need the value at the same time
 - What value should be seen
- Consistency — All threads see the change of data in the same order
 - When the memory operation should be done

Simple cache coherency protocol

- Snooping protocol
 - Each processor broadcasts / listens to cache misses
- State associate with each block (cacheline)
 - Invalid
 - The data in the current block is invalid
 - Shared
 - The processor can read the data
 - The data may also exist on other processors
 - Exclusive
 - The processor has full permission on the data
 - The processor is the only one that has up-to-date data

Coherent way-associative cache

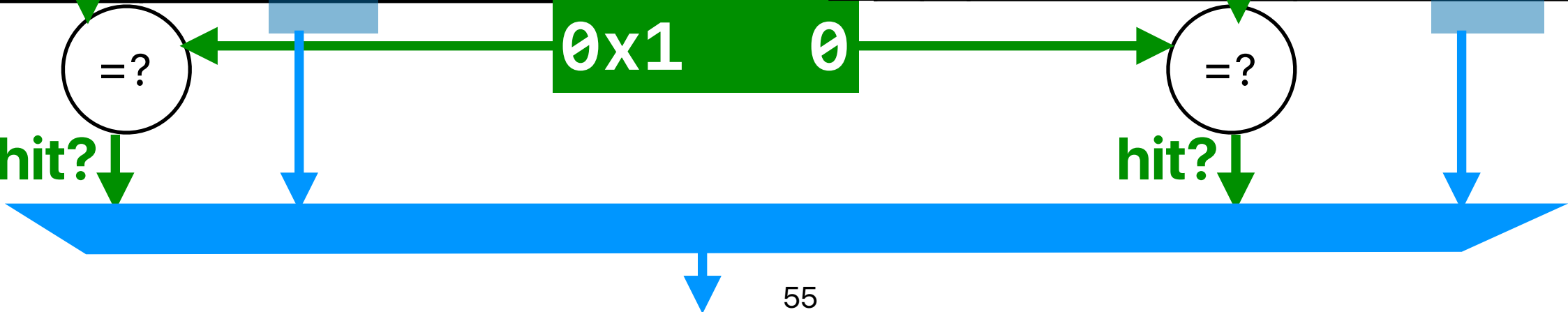
memory address: 0x0

memory address: 0b00001000000100100

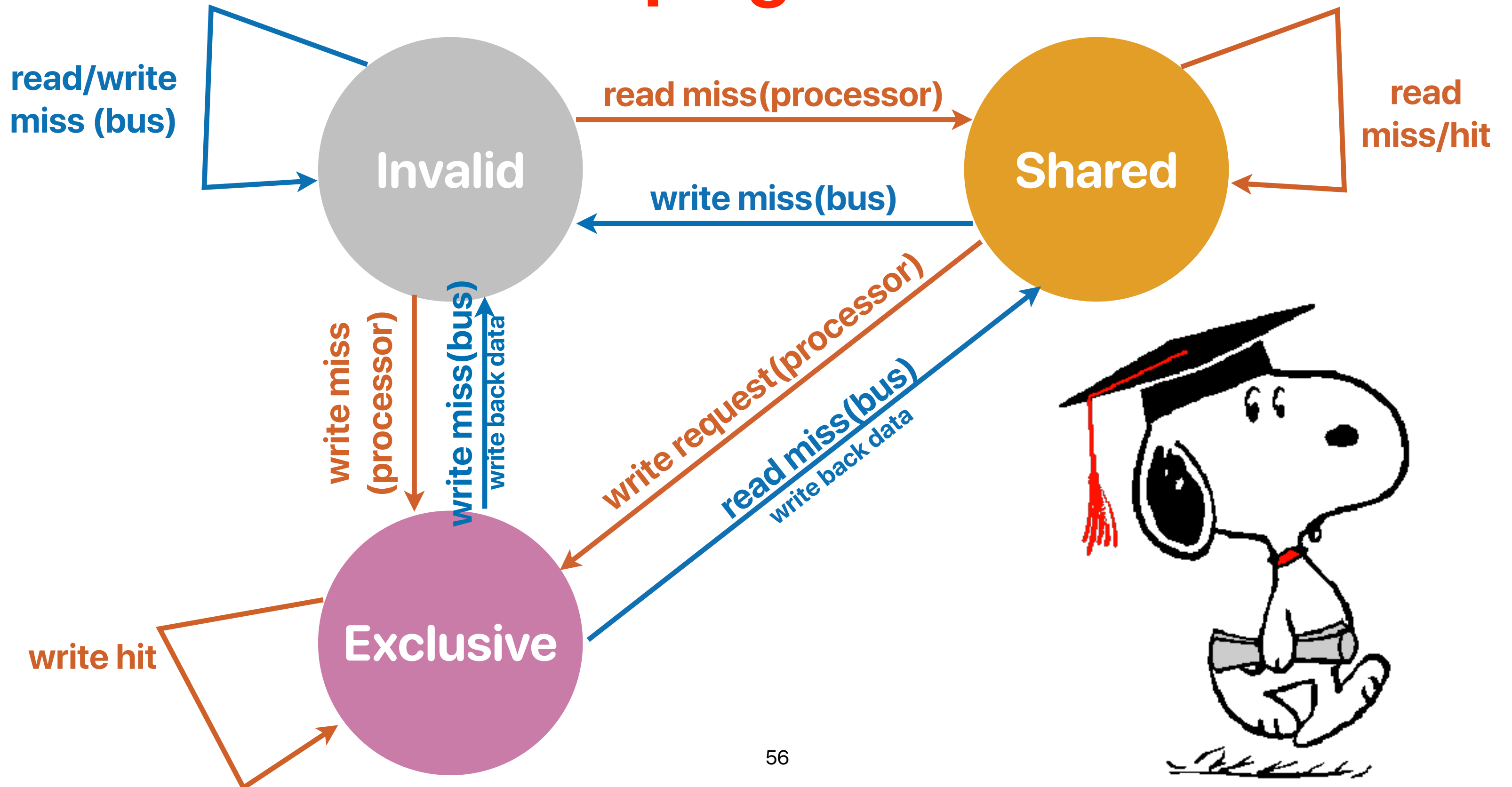
8 tag 2 set 4 block
index offset

States	D	tag	data
01	1	0x29	IIJJKKLLMMNNOOPP
01	1	0xDE	QQRRSSTTUUVVWXX
01	0	0x10	YYZZAABBCCDDEEFF
00	1	0x8A	AABBCCDDEEGGFFHH
10	1	0x60	IIJJKKLLMMNNOOPP
10	1	0x70	QQRRSSTTUUVVWXX
10	1	0x10	QQRRSSTTUUVVWXX
10	1	0x11	YYZZAABBCCDDEEFF

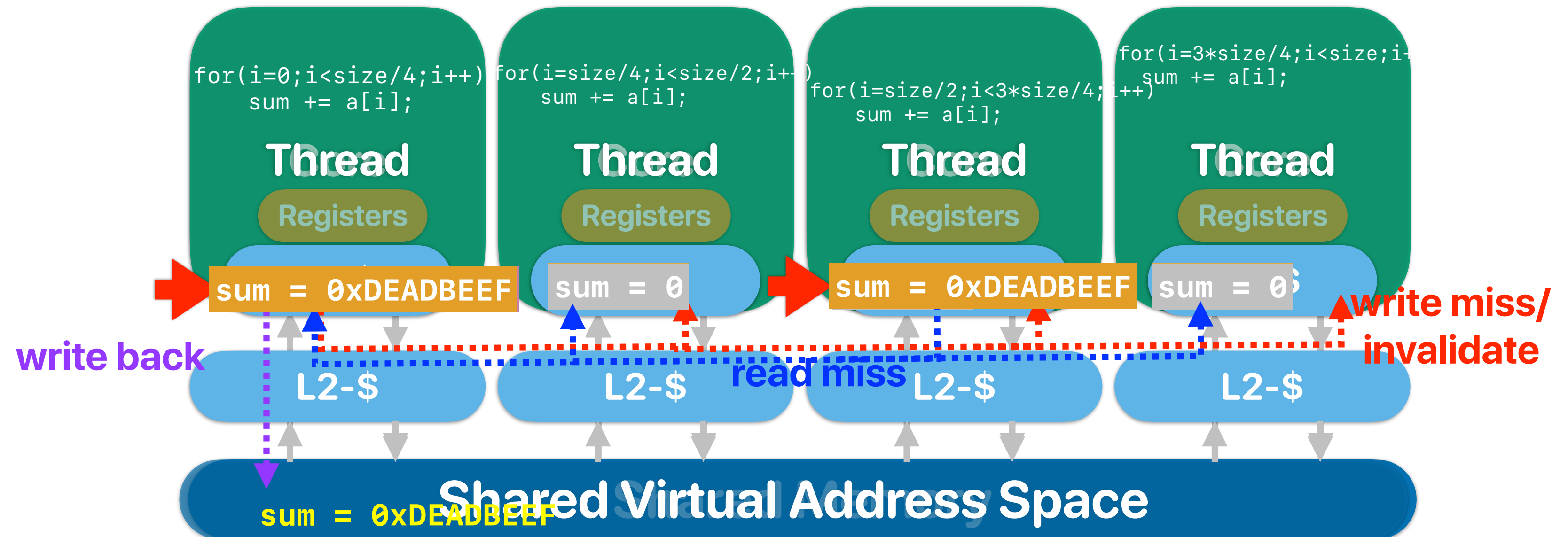
States	D	tag	data
01	1	0x00	AABBCCDDEEGGFFHH
01	1	0x10	IIJJKKLLMMNNOOPP
01	0	0xA1	QQRRSSTTUUVVWXX
00	1	0x10	YYZZAABBCCDDEEFF
10	1	0x31	AABBCCDDEEGGFFHH
10	1	0x45	IIJJKKLLMMNNOOPP
10	1	0x41	QQRRSSTTUUVVWXX
10	1	0x68	YYZZAABBCCDDEEFF



Snooping Protocol



What happens when we write in coherent caches?



Observer

thread 1

```
int loop;

int main()
{
    pthread_t thread;
    loop = 1;

    pthread_create(&thread, NULL,
modifyloop, NULL);
    while(loop == 1)
    {
        continue;
    }
    pthread_join(thread, NULL);
    fprintf(stderr, "User input: %d\n",
```

thread 2

```
void* modifyloop(void *x)
{
    sleep(1);
    printf("Please input a number:\n");
    scanf("%d",&loop);
    return NULL;
}
```

Observer

prevents the compiler from putting the variable "loop" in the "register"

thread 1

thread 2

```
volatile int loop;

int main()
{
    pthread_t thread;
    loop = 1;

    pthread_create(&thread, NULL,
modifyloop, NULL);
    while(loop == 1)
    {
        continue;
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    scanf("%d",&loop);
    return NULL;
}
```

Announcements

- **Reading Quiz 8 (last reading quiz)** due **next Tuesday** before the lecture
- **Assignment 4 is released** due this Saturday
- **Assignment 5 is released** due 9/5/2024

Computer Science & Engineering

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